pfi

# Python for Fixed Income

COURSE 4 · MACHINE LEARNING

# ROC Curves and Choosing a Threshold

WEEK 2 · DAY 3

Created by **Jeff Lenamon**

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

# Where day 7 left off

### Day 7: threshold 0.5

**324 of 1,327**

defaults caught, 6,000 validation rows

Recall 0.2442, precision 0.7248. Nothing about 0.5 says it is the place to stop.

### Real data

UCI card file: accounts in Taiwan, 2005. Not US credit today.

### People's attributes

SEX, EDUCATION, MARRIAGE, AGE: never features.

### 60 / 20 / 20

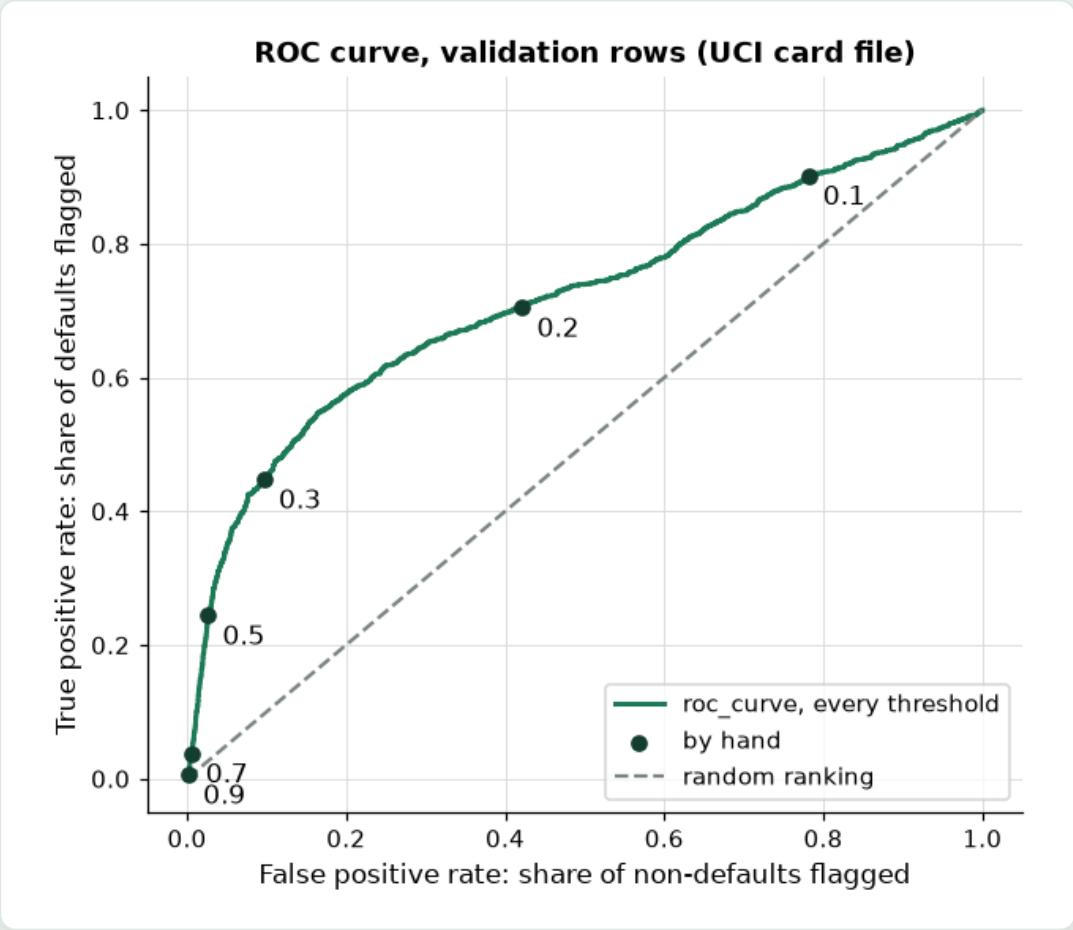
Training fits, validation chooses, test reports once (day 10).

### Names

Model probability of default in this file; flagged at threshold t.

**Every threshold and target today is a teaching convention, not what a lender does or should do.**

# Every threshold at once: the ROC curve

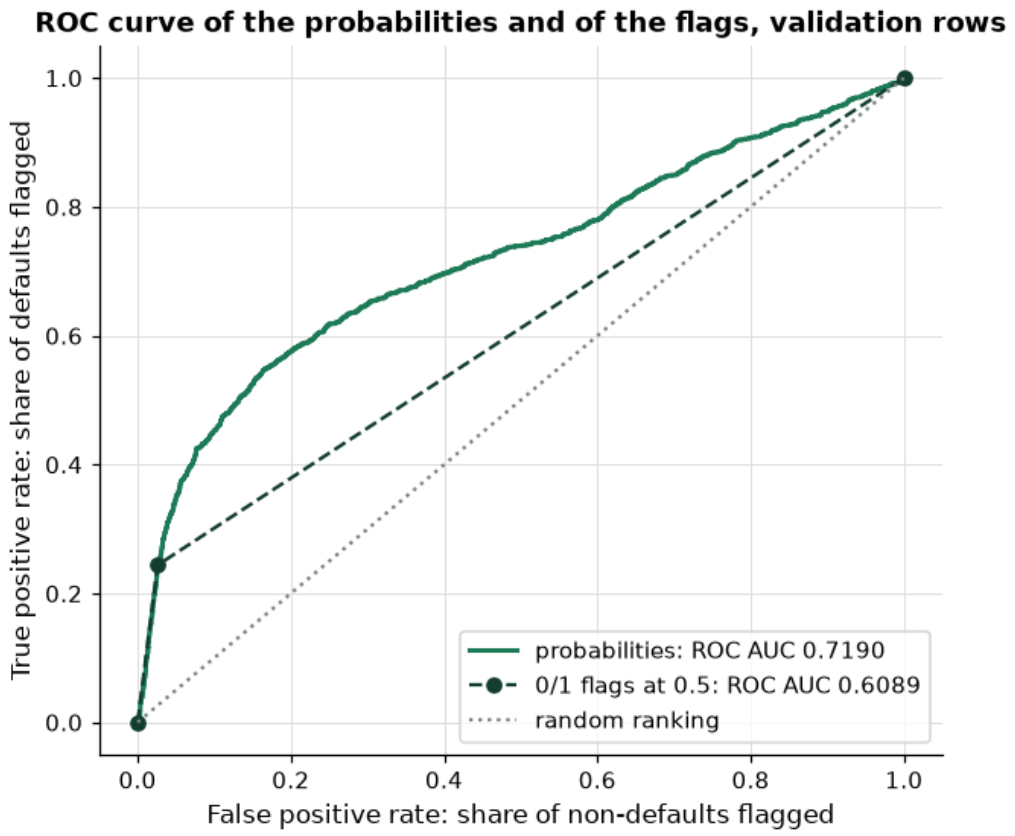



| Rate | Formula | Out of |
|---|---|---|
| True positive (recall) | TP / (TP + FN) | 1,327 defaults |
| False positive | FP / (FP + TN) | 4,673 others |

| Threshold | Flagged | TPR | FPR |
|---|---|---|---|
| 0.5 | 447 | 0.2442 | 0.0263 |
| 0.3 | 1,047 | 0.4476 | 0.0969 |
| 0.2 | 2,905 | 0.7046 | 0.4216 |
| 0.1 | 4,848 | 0.9020 | 0.7813 |

**Up the curve is a lower threshold. The diagonal is a score with no information.**

# AUC as a ranking measure




**0.7190**

ROC AUC, probabilities

**0.6089**

the 0/1 flags at 0.5

One account that defaulted, one that did not: AUC is the chance the model ranks the defaulted one higher. Here, 71.90 percent of 6,201,071 pairs.

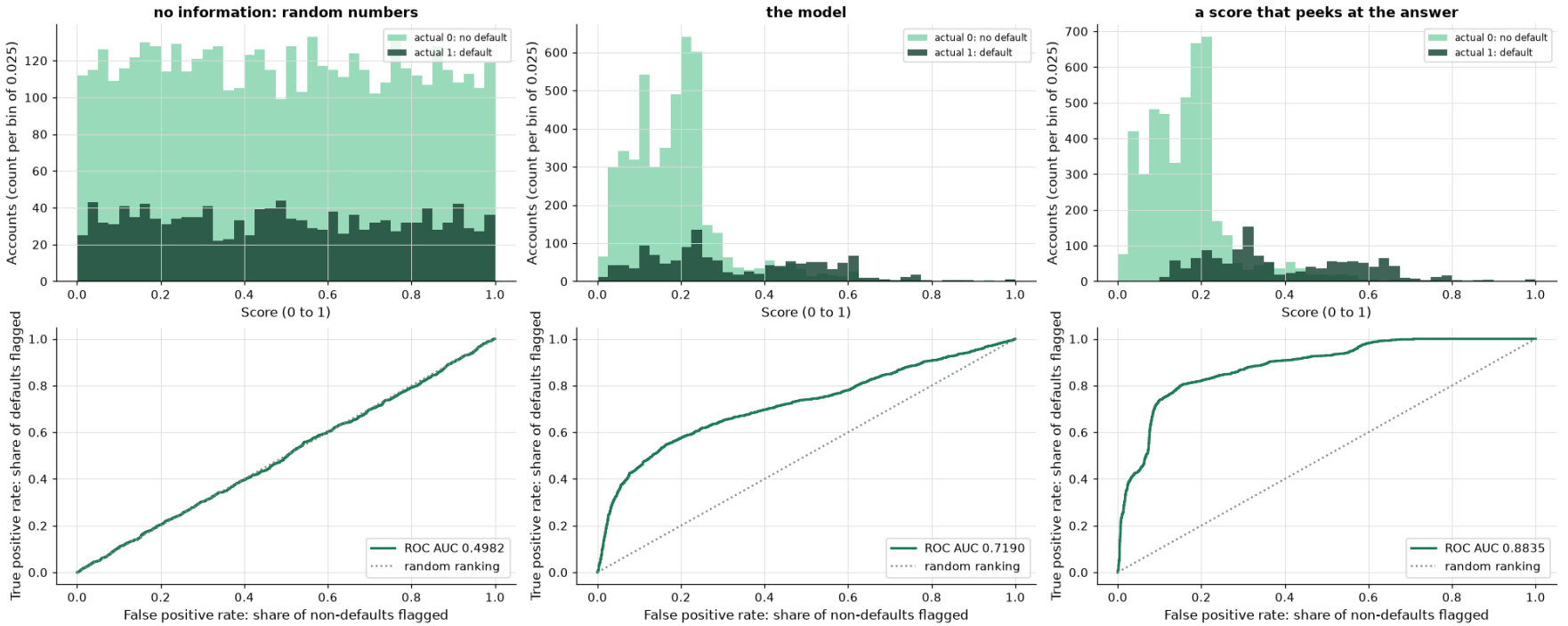
### What AUC does not say

Not an accuracy, not a default rate

Nothing about any one threshold

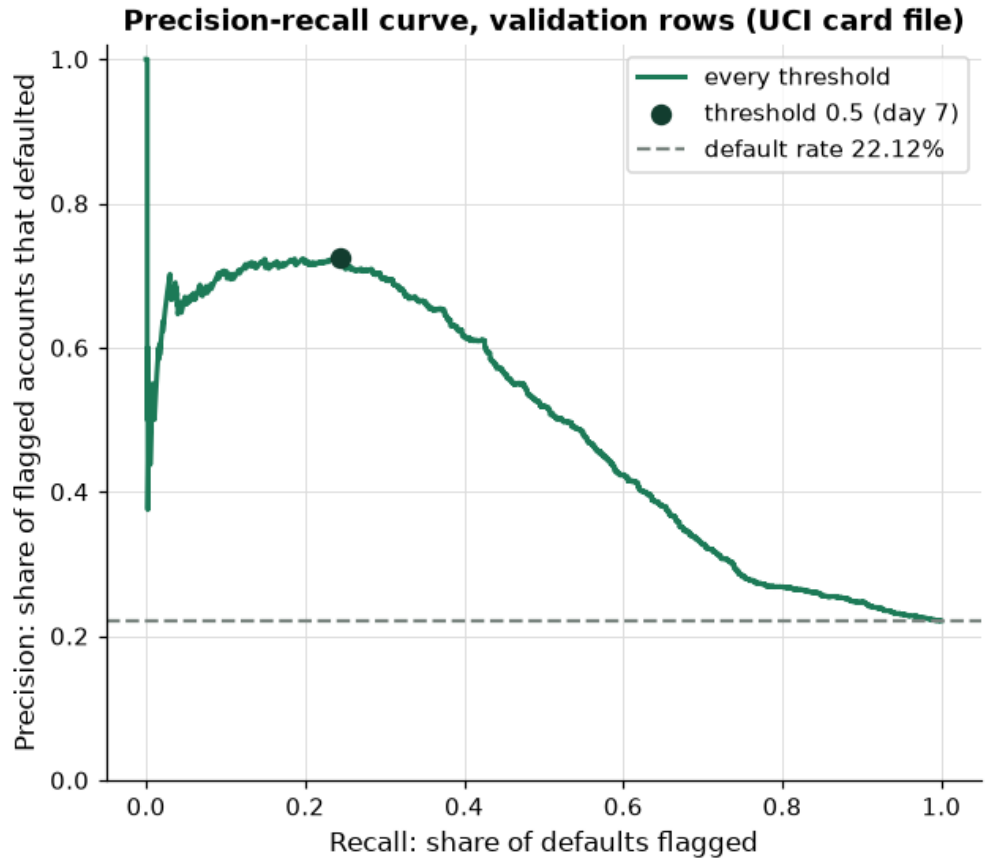
Not whether a 0.3 means three in ten: squared, 0.7190

# AUC as overlap




**AUC 0.4982 for random numbers, 0.7190 for the model, 0.8835 for a score that has seen the answer: the less the groups overlap, the higher the curve.**

# The precision-recall curve




**0.4946**

average precision

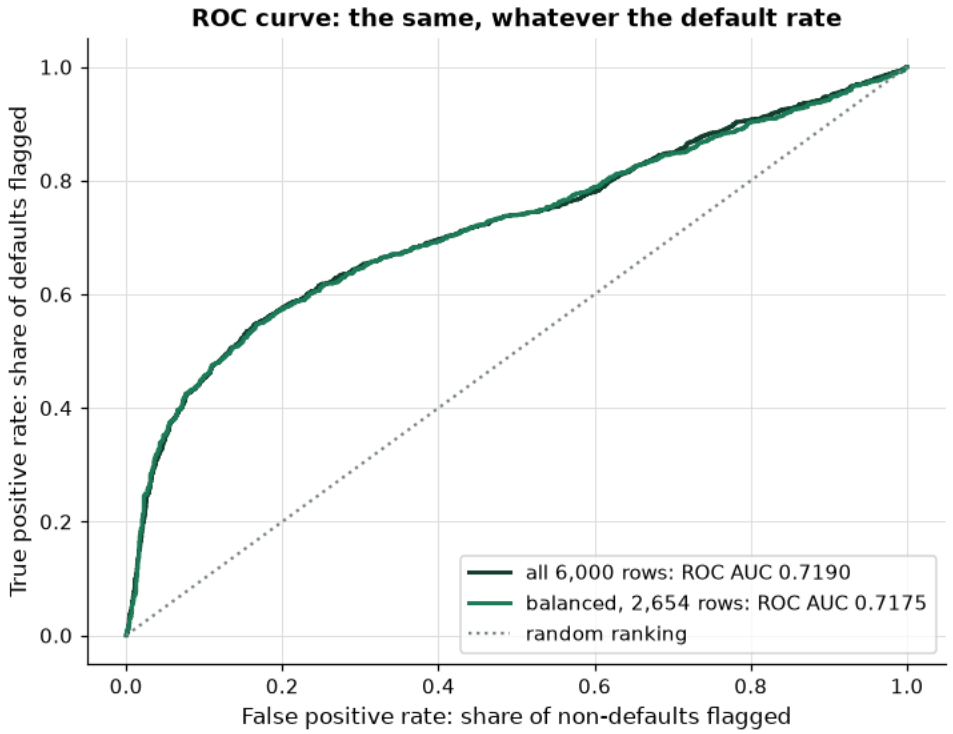
**22.12%**

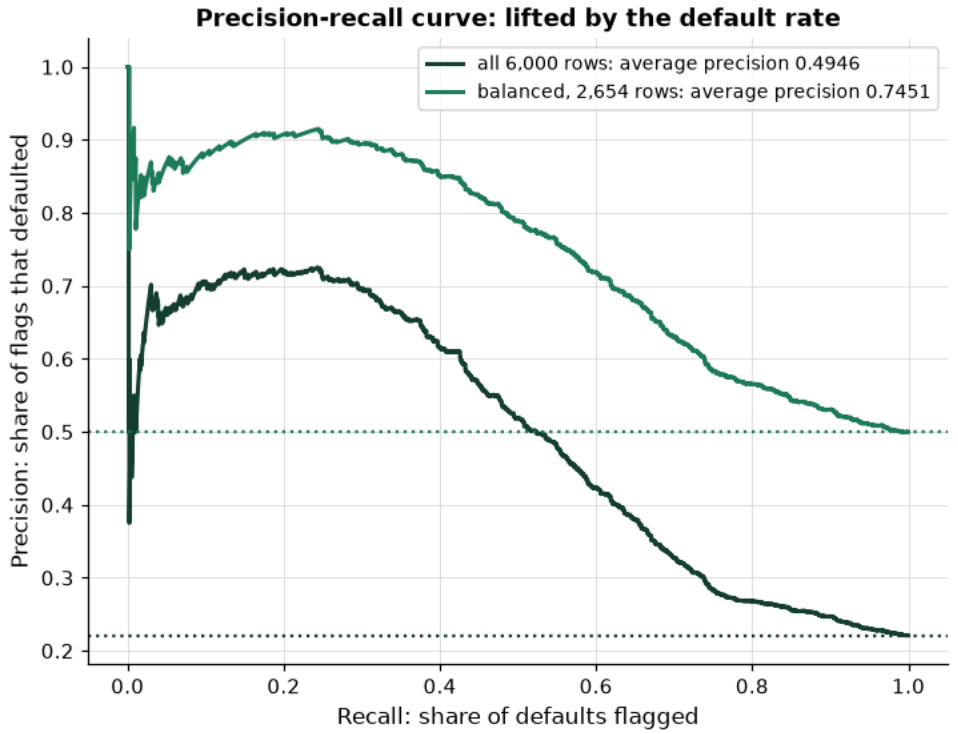
the floor: the default rate

Precision divides by the flagged accounts: of those flagged, how many defaulted? The false positive rate divides by every non-default.

**Average precision is a step sum: each rise in recall times the precision there. Not a trapezoid.**

# Does the default rate change the curves?




**ROC AUC 0.7190 and 0.7175; average precision 0.4946 and 0.7451. Quote average precision with the default rate beside it.**

# Choosing a threshold on validation rows

### The course's rule: mlkit.threshold_for_recall

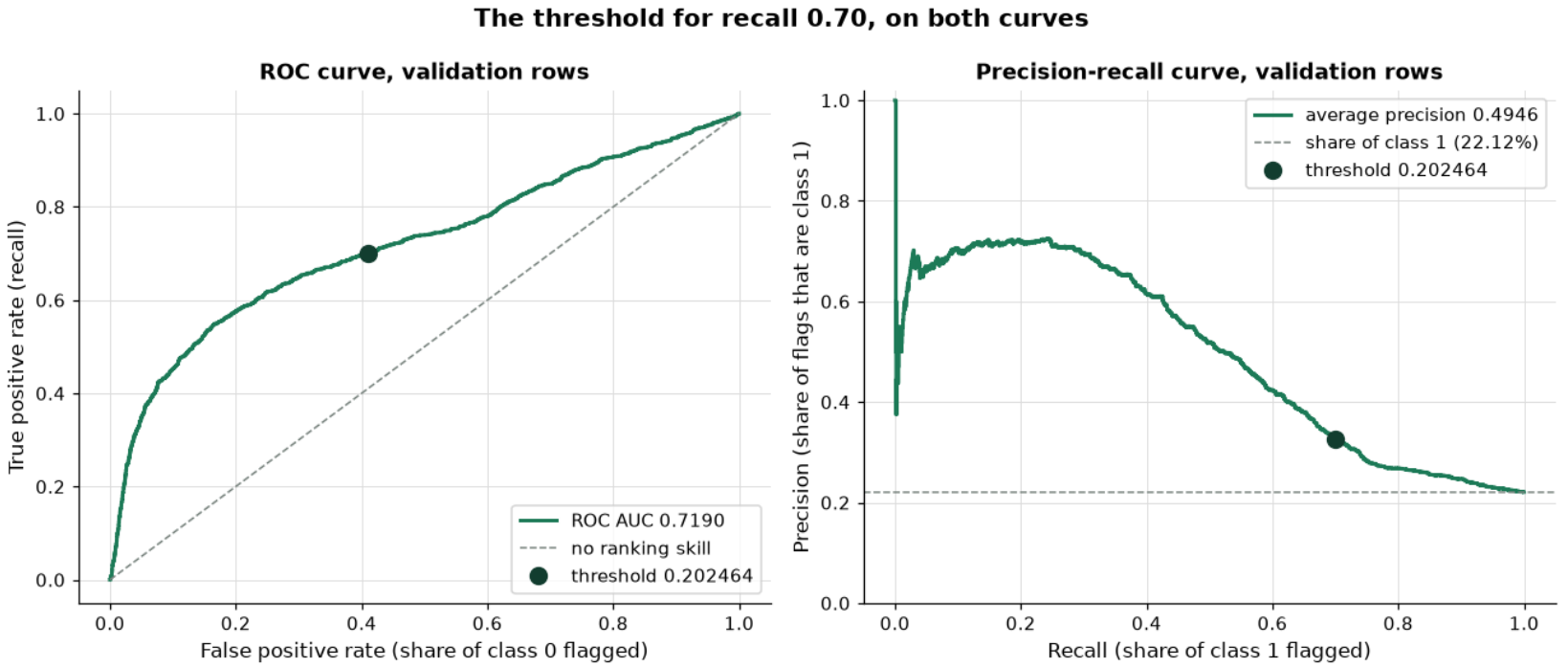
If the desk asked for recall of 70 percent: the highest threshold with validation recall at least 0.70.

| Rule | Threshold | Flagged | Recall | Precision | F1 |
|---|---|---|---|---|---|
| Day 7: 0.5 | 0.5000 | 447 | 0.2442 | 0.7248 | 0.3653 |
| Round 0.2 | 0.2000 | 2,905 | 0.7046 | 0.3219 | 0.4419 |
| Recall at least 0.70 | 0.2025 | 2,843 | 0.7001 | 0.3268 | 0.4456 |
| Youden's J | 0.2475 | 1,507 | 0.5494 | 0.4837 | 0.5145 |

Youden's J: where TPR minus FPR is largest (0.3829). Accuracy of flagging nothing: 0.7788; never used to choose.

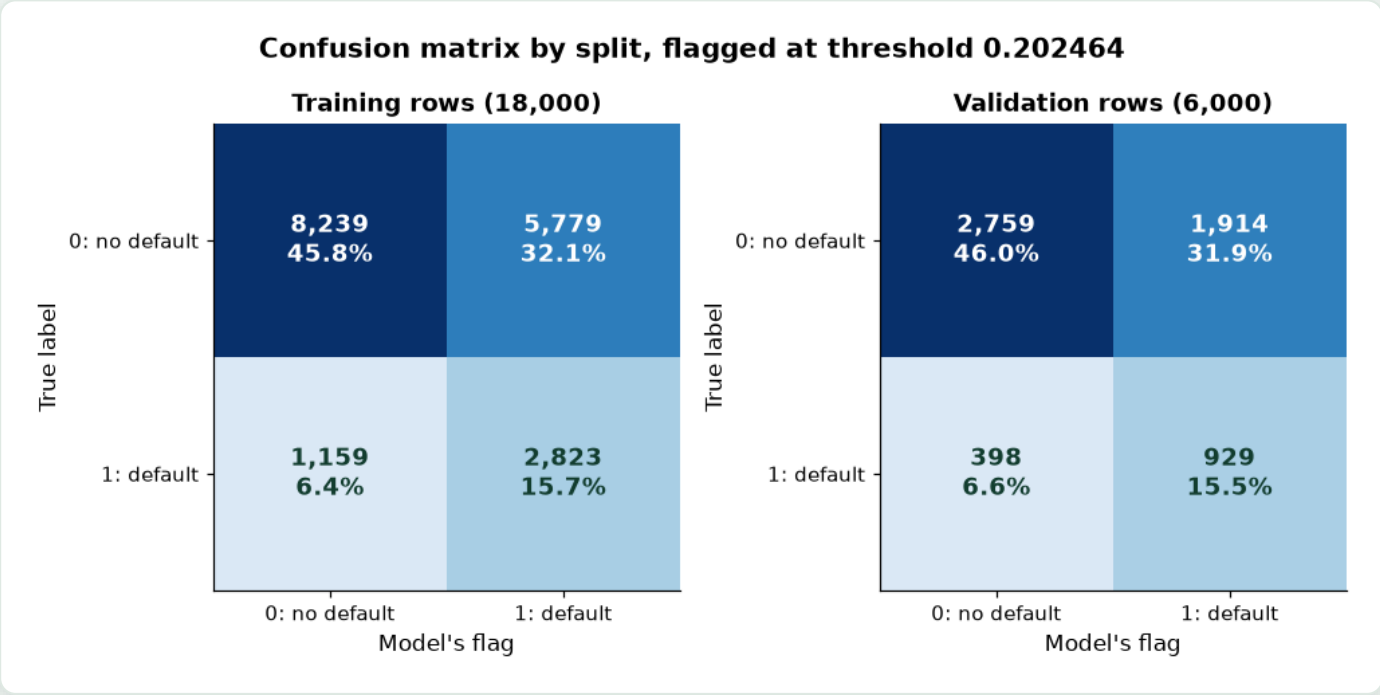
**A teaching convention, not a lending standard. No threshold here is what a lender does or should do.**

# The chosen threshold on both curves




**One threshold, one point on each curve: recall 0.7001, precision 0.3268. About one flagged account in three defaulted.**

# The chosen threshold, training and validation rows

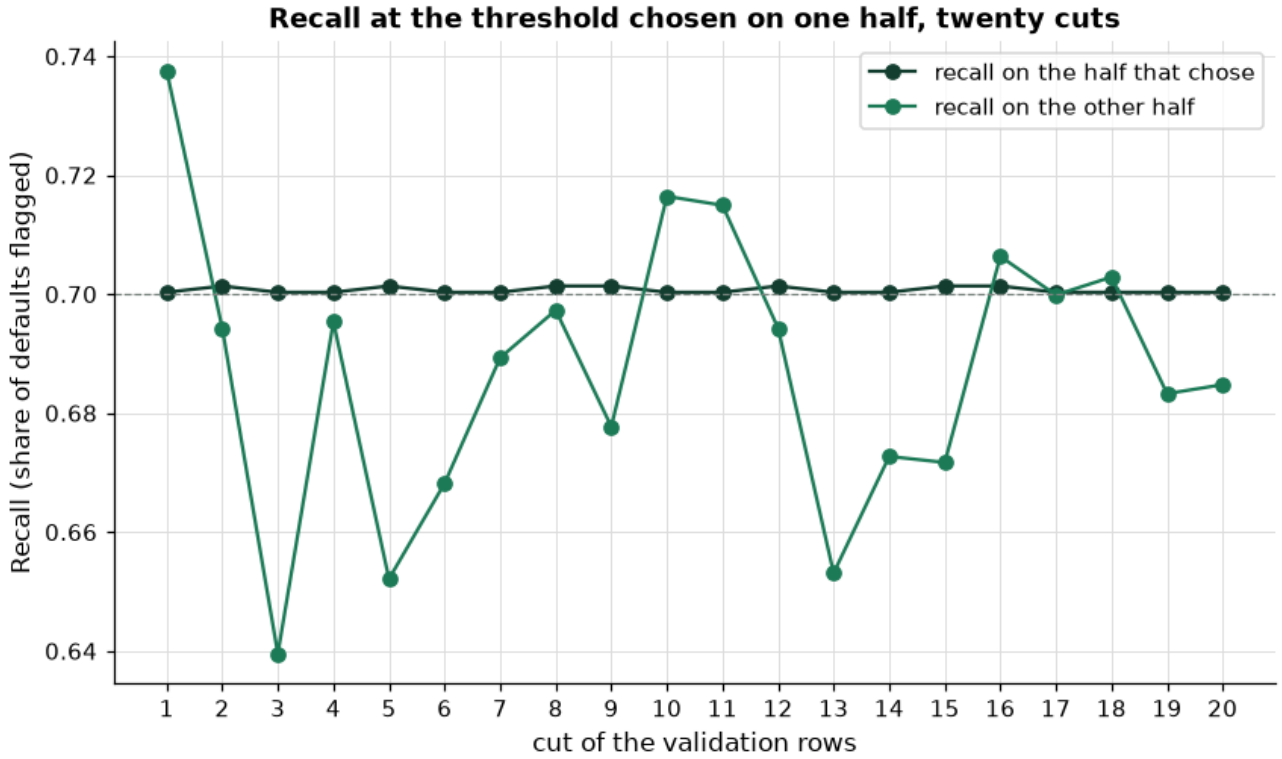



| Rows | Recall | Precision | F1 |
|---|---|---|---|
| training | 0.7089 | 0.3282 | 0.4487 |
| validation | 0.7001 | 0.3268 | 0.4456 |

Read the panels as shares: the splits differ in size. The two rows are close.

**The validation rows chose this threshold, so their recall is not the number to report.**

# Why the test rows are not used to choose




**0.7003 to 0.7014**

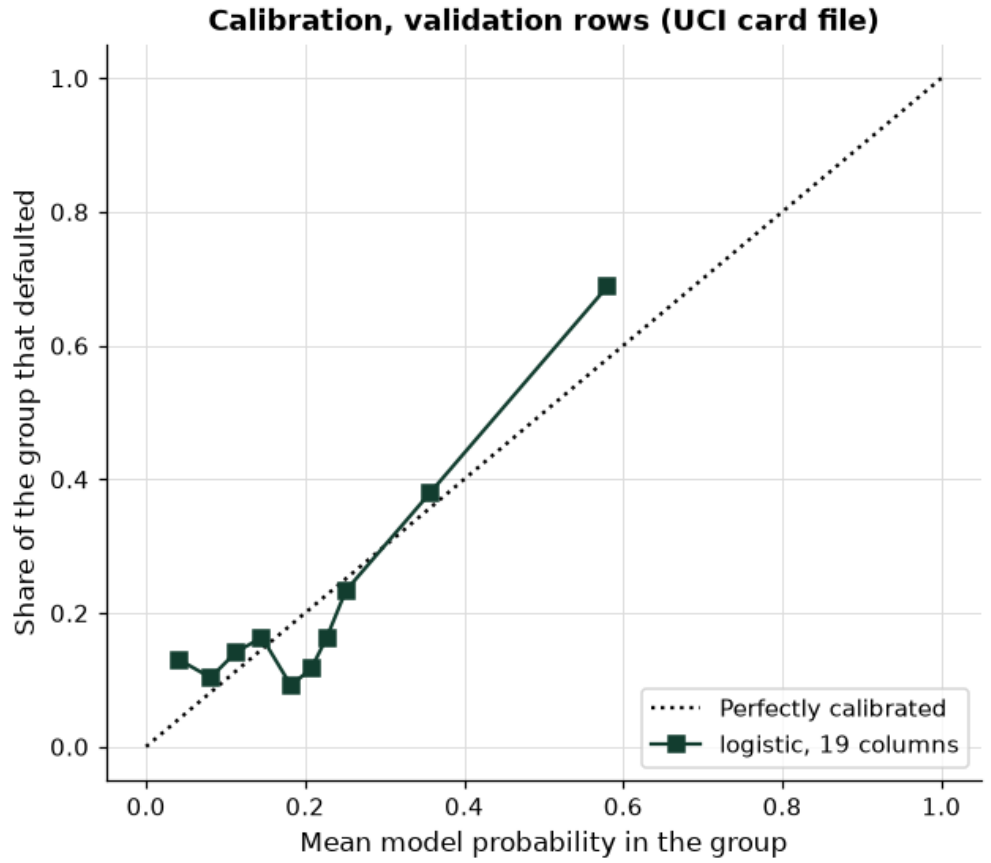
the half that chose

**0.6395 to 0.7376**

the other half: below 0.70 in 15 of 20

**Chosen on the test rows, the reported recall would be 0.70 by construction. Day 10 opens them once.**

# Calibration: does a 0.3 mean three in ten?




| Group | Mean probability | Defaulted |
|---|---|---|
| Lowest | 0.0416 | 13.00% |
| Nearest 0.3 | 0.2512 | 23.33% |
| Top | 0.5800 | 68.83% |

Ten groups of about 600 accounts, sorted by probability. On the diagonal, a probability means what it says.

**It ranks better than chance; its probabilities are not uniformly reliable as numbers. Recalibration: Course 5.**

# Excel side by side

**Excel side by side (tested in Excel)**

| | |
|---|---|
| `=COUNTIFS(A2:A6001,1,B2:B6001,">="&D2)  (row 2)` | TP 324 at 0.5 |
| `=F2/COUNTIF(A2:A6001,1)` | recall 0.2442 |
| `=SORT(UNIQUE($B$2:$B$6001),,-1)  (M3)` | 5,882 thresholds |
| `=COUNTIFS($A$2:$A$6001,1,$B$2:$B$6001,">="&M3#)/COUNTIF($A$2:$A$6001,1)  (O3)` | TPR, spilled |
| `=SUMPRODUCT((N3:N5884-N2:N5883),(O3:O5884+O2:O5883)/2)` | AUC 0.7190 |
| `=MAXIFS(M3#,O3#,">=0.7")` | rule 0.2025 |
| `=INDEX(M3#,MATCH(MAX(O3#-N3#),O3#-N3#,0))` | Youden 0.2475 |
| `=SUMPRODUCT((O3:O5884-O2:O5883),P3:P5884)` | AP 0.4946 |

**Excel has no ROC or AUC function: the probabilities come from Python; every step after them is a formula.**

# Save the day: the history of the choices

```
git log --oneline -- results.csv
  Experiment log: Youden's J threshold beside it, for comparison
  Experiment log: the threshold for recall 0.70, chosen on validation rows
  Experiment log: the logistic model at day 7's threshold 0.5
```

```
model,rows,roc_auc,average_precision,threshold,accuracy,precision,recall,f1
logistic_scaled at fixed 0.5,validation,0.7190,0.4946,0.5000,0.8123,0.7248,0.2442,0.3653
logistic_scaled at recall at least 0.70,validation,0.7190,0.4946,0.2025,0.6147,0.3268,0.7001,0.4456
logistic_scaled at Youden J,validation,0.7190,0.4946,0.2475,0.7707,0.4837,0.5494,0.5145
```

One row per choice, one commit per row. The two dashes keep only the commits that changed results.csv.

**Why the threshold is what it is: the log shows the order of the choices and the commit behind each.**

# The AI check

The docstring is the prompt. The code that comes back runs, and has one modelling bug.

1

## Docstring

Which rows fit, which rows report, and the units: what you asked for.

2

## Code back

Written for the course in the style of an AI answer. Read it.

3

## Test first

Assert what the docstring promises. Run it: it fails.

4

## Fix

Find the rule it broke, fix it, run the test again.

**Today: the threshold is chosen on the test rows, with a comment that says the model never saw them.**

*Code that runs is not the same as code that is right.*

# What you can do now

- The ROC curve: every threshold at once, each point a confusion matrix
- AUC 0.7190: pairs ranked the right way, nothing about one threshold
- Average precision 0.4946: a step sum over the precision-recall curve
- Recall at least 0.70 on validation rows: threshold 0.2025, a convention
- The rows that choose cannot report; calibration checks the numbers

**Week 2, Day 4** Decision trees: a model made of questions, on the same validation rows